1. Objectives

The goal is to ensure that the core functionality of the SauceDemo e-commerce website, including login, product browsing, cart management, and checkout, meets the expected functionality and usability standards.

• Verify authentication and login validation across all specified user types, using the expected outcome for each account.

• Ensure accurate addition and removal of cart items, correct cart badge counts, and successful checkout with valid information.

• Verify consistent functional behavior, layout, and navigation across the latest versions of Chrome, Firefox, and Edge.

2. Approach

The test strategy combines manual and automated testing.

• **Manual testing:** UI validation, usability checks, exploratory testing, and functional verification.

• **Automated testing:** Selenium WebDriver regression tests for critical workflows, including login and checkout.

• **Test types:** Functional, UI, input validation, end-to-end, and cross-browser compatibility testing.

• **Input coverage:** Valid and invalid credentials, empty login fields, required checkout fields, and relevant cart states.

Execution and reporting

Execute tests against defined expected results. Record actual results as Pass, Fail, Blocked, or Not Run. Log reproducible defects in Jira and include coverage, execution counts, and unresolved issues in the final summary.

Performance and load testing, security testing, and real payment processing are excluded. Functional testing of performance_glitch_user remains in scope.

3. Scope

In Scope

• **User authentication:** Login with standard_user, locked_out_user, problem_user, performance_glitch_user, error_user, and visual_user.

• **Login validations:** Invalid credentials, empty fields, and the locked-out user error message.

• **Logout:** Logout through the side menu.

• **Products page:** Product details display and sorting by name and price.

• **Product details page:** Open a product and return to the products page.

• **Shopping cart:** Add and remove items, verify the badge count, and validate cart persistence.

• **Checkout workflow:** Your Information, Overview with item total, tax, and total, and Complete.

• **Side menu:** All Items, About navigation, Logout, and Reset App State.

• **UI and usability:** Layout, images, buttons, and navigation.

• **Cross-browser compatibility:** Chrome, Firefox, and Edge.

Out of Scope

• Performance and load testing.

• Security and penetration testing.

• Real payment processing, as the website is a demo.

• Backend, database, and API testing.

• External destination content, including the About page and footer social media links.

• Safari and older browser versions.

• Native mobile applications.

4. Test Environment

Item	Details
Application URL	https://www.saucedemo.com
Environment Type	Remote (public demo website)
Operating System	Windows 10/11
Browsers	Google Chrome, Mozilla Firefox, Microsoft Edge (latest versions)
Device	Desktop/Laptop (1920 x 1080)
Network	Stable Wi-Fi / broadband internet connection
Test Data	Usernames: standard_user, locked_out_user, problem_user, performance_glitch_user, error_user, visual_user. Password for all users: secret_sauce. Checkout: any valid First Name, Last Name, and Postal Code.
Automation Tools	Selenium WebDriver
Test Management & Bug Tracking	Jira

5. Dependencies

- **Website Availability:** SauceDemo must be accessible during testing.
- **Internet Connection:** A stable internet connection is required.
- **Test Accounts:** Valid test credentials are needed to test login behavior.
- **Browsers:** Supported versions of Chrome, Firefox, and Edge must be available.
- **Test Data:** Required product and checkout data must be prepared before execution.

6. Test Deliverables

The following deliverables will be prepared during manual and automated testing of SauceDemo:

Deliverable	Content
Test Scenarios	Main features to be tested, including Login, Product Browsing, Shopping Cart, and Checkout.
Test Cases	Detailed steps, preconditions, test data, and expected results for each scenario.
Test Execution Results	Actual results and execution status for each case: Pass, Fail, Blocked, or Not Run.
Bug Reports	Defect descriptions, severity, steps to reproduce, expected and actual results, and supporting screenshots.
Automation Scripts & Results	Selenium WebDriver regression scripts and execution reports for selected critical workflows.
Test Summary Report	Final coverage, passed and failed case counts, blocked or unexecuted tests, identified bugs, and unresolved issues.

Frequency

• Test scenarios and test cases will be prepared before testing.

• Test results, automation results, and bug reports will be updated during testing.

• The test summary report will be prepared at the end of the test cycle.

Distribution

All test deliverables will be shared with QA team members through a shared folder. Defects will be tracked in Jira.

Formats

Use shared spreadsheets or documents for scenarios, cases, and execution records; Jira for defects; Selenium script files and test runner reports for automation; and a PDF or document for the final summary.

7. Risk Management

The following risks may affect SauceDemo testing. Each risk will be monitored, and mitigation actions will be taken as needed.

Risk	Impact	Mitigation
Website unavailable	Testing may be delayed or blocked.	Check availability before execution, record blocked tests, and resume when the website becomes accessible.
Unstable internet connection	Tests may fail or produce unreliable results.	Use a stable connection and rerun affected tests after connectivity is restored.
Unexpected behavior from special user accounts	Intended account behavior may be reported as a defect.	Define expected results for each user type and compare affected flows with standard_user before reporting defects.
Cart or session data remains between tests	Previous test data may affect later results.	Establish a clean starting state before each independent test. Preserve state intentionally for cart persistence tests.
Limited testing time or tester availability	Some planned tests may remain incomplete.	Prioritize login, cart management, and checkout; document any untested areas in the summary report.
Browser differences or Selenium failures	Tests may fail on specific browsers or produce misleading automation results.	Record browser versions, investigate failures, and manually verify affected flows.
Website changes during testing	Test cases or automation scripts may become outdated.	Review affected cases, update scripts, and rerun relevant regression tests.

Monitoring: Risks and blocked tests will be reviewed daily during test execution. Any impact on scope, coverage, or completion dates will be communicated to the QA team and recorded in the test summary report.

SAUCEDEMO WEBSITE

Testing Plan – Schedule, Roles & Responsibilities

8. Test Schedule

The following is a proposed 10-working-day testing timeline for the SauceDemo website (https://www.saucedemo.com/). It covers the main user journey: login, product browsing and sorting, adding/removing items from the cart, checkout, order completion, and logout. The timeline can be adjusted depending on team size and the availability of a stable test build.

Phase / Activity	Main Tasks	Duration	Milestone / Deadline
1. Planning & setup	Confirm scope, review requirements, identify risks, prepare test environment and test accounts.	Day 1	End of Day 1: Test plan and scope approved.
2. Test design	Write test scenarios and test cases for login, inventory, sorting, cart, checkout, and logout.	Days 2–3	End of Day 3: Test cases reviewed and ready.
3. Manual testing – functional	Execute positive and negative tests; verify validation messages, product details, cart totals, and checkout flow.	Days 4–5	End of Day 5: First execution cycle completed; defects logged.
4. Defect fixing & retesting	Report and prioritize defects, verify fixes, and perform regression checks on affected areas.	Days 6–7	End of Day 7: High-priority defects retested.
5. Automation testing	Automate stable, high-value regression cases such as login, adding an item to cart, and checkout using an agreed tool (e.g., Selenium).	Days 8–9	End of Day 9: Initial automated regression suite executed.
6. Final testing & reporting	Run final smoke/regression tests, summarize results, document open issues, and prepare the test completion report.	Day 10	End of Day 10: Test summary and sign-off recommendation delivered.

Key Schedule Notes

• **Daily tracking:** The QA team will update test execution status and report blockers at the end of each working day.

• **Entry criteria:** The test environment is accessible, the agreed scope is available, and test accounts can be used.

• **Exit criteria:** Planned test cases have been executed, critical/high-priority defects are resolved or formally accepted, and test results are documented.

• **Automation note:** Automation starts after the main flows are understood and reasonably stable; manual testing remains necessary for exploratory checks and usability observations.

9. Roles and Responsibilities

The team roles below are suggested for a small QA project. One person may cover more than one role if the team is small. All team members should share clear updates and record decisions and defects.

Role	Responsibilities
QA / Test Lead	Defines the test scope and strategy; prepares the schedule; assigns tasks; monitors progress; reviews test cases and defect priorities; communicates risks; and prepares the final test summary.
Manual Tester	Designs and executes manual test cases; checks login, inventory, sorting, cart, checkout, and logout; records actual results; captures evidence; and reports defects with clear reproduction steps.
Automation Tester	Selects suitable stable regression scenarios; develops and maintains automated scripts; runs the suite; investigates failures; and shares execution reports. Coordinates with the manual tester to avoid gaps.

Role	Responsibilities
Developer	Reviews reported defects, investigates root causes, fixes confirmed issues, provides updated builds, and supports the QA team with technical clarification when needed.
Project Coordinator	Tracks deadlines and dependencies; arranges meetings; shares status updates; and ensures that decisions, approvals, and release communications reach the relevant team members.

10. Communication Channels & Protocols

Channel	Purpose / Protocol
Team chat (e.g., Microsoft Teams or WhatsApp)	Daily updates, quick questions, blockers, and urgent notifications. Keep messages concise and mention the relevant task or defect ID.
Task board (e.g., Trello or Jira)	Track tasks using clear statuses such as To Do, In Progress, Blocked, and Done. Assign an owner and due date to each task.
Defect tracker	Each defect should include a title, environment, steps to reproduce, expected result, actual result, severity/priority, and supporting evidence.
Daily check-in	Hold a short daily meeting to share completed work, planned work, and blockers. Escalate any issue that may affect a milestone to the QA Lead.

Assumption: This is a proposed academic testing plan, not a record of tests already performed. The 10-day schedule uses relative working days because no project start date or team member names were provided.